

COURSE 4- ADVANCED LINUX SYSTEM ADMINISTRATION

SPRINT 2 - CONFIGURE NETWORK SERVICES

PRACTICE 1- CONFIGURE THE DHCP SERVER

Problem Scenario

SCR Corporation is a leading technology organization that provides network support services to its clients globally. You are hired as a network administrator and are responsible for setting up the network infrastructure for your organization's clients. As a network administrator, you need to perform the practices mentioned

PRACTICE:

One of the clients of SCR Corporation has more than 100 systems. Configure the DHCP Server One of the clients of SCR Corporation has more than 100 systems. Configuring IP address manually on each system will be a time-consuming task and might result in human errors. Therefore, as a network administrator, you are required to:

- Configure a DHCP server and systems to receive IP addresses automatically from the server.
- Verify that the DHCP process works properly.

Tasks:

Configure the DHCP Server You have been provided three machines in the lab: G1, M1, and M2.

- Install and configure the DHCP service on the G1 vagrant machine.
- Configure the DHCP scope to include IP addresses in the range:
192.168.1.1-192.168.1.150
- Configure the vagrant boxes M1 and M2 to get the IP address from the DHCP server.

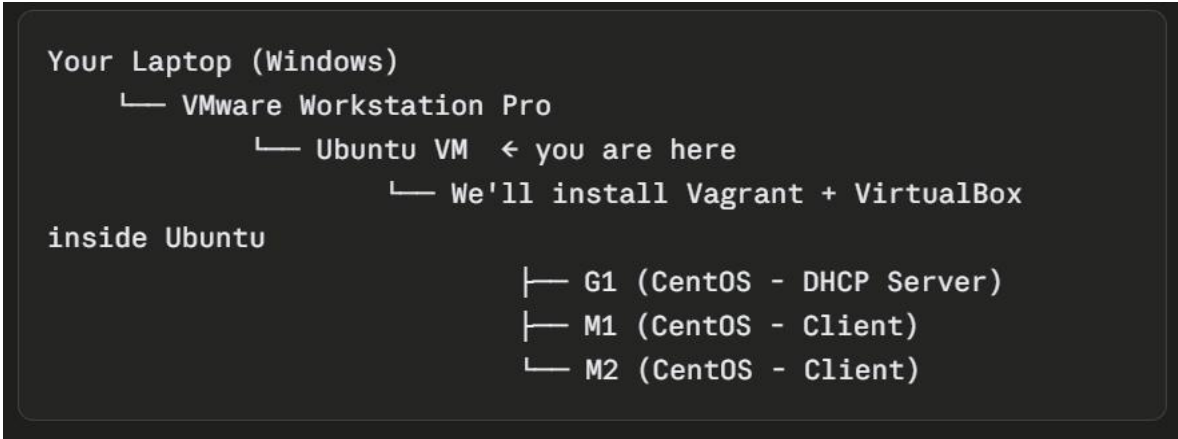
- Verify that the IP addresses on machines M1 and M2 were automatically provided by the DHCP server.

Notes: The method you use to configure M1 and M2 will be the same for configuring the remaining 98 systems.

In my System (lab setup) - UBUNTU 24.04 DESKTOP VERSION IN VMWARE WORKSTATION PRO 17

Ubuntu running in VMware Workstation Pro on the laptop. Let's set everything up from scratch inside the Ubuntu VM.

SETUP PLAN:



NEED TO DOWNLOAD VAGRANT AND VITUALBOX IN UBUNTU VIRTUAL MACHINE

Lab Overview

Scenario 1	LXC Containers — G1 (DHCP + DNS Server), M1 & M2 (Clients)
Scenario 2	Vagrant + VirtualBox — G1 (DHCP Server), M1 & M2 (Clients)
OS Used	CentOS 9-Stream (LXC) / CentOS 8 (Vagrant)
Network (S1)	10.201.2.0/24 (LXC bridge network)
Network (S2)	192.168.10.0/24 (VirtualBox host-only)
Status	scenario completed and verified successfully

Topics included

- DHCP server installation
- DHCP configuration
- DHCP IP range/scope configuration
- Client configuration
- DHCP verification/testing

Step 1: Check the version of Ubuntu & update Ubuntu First

Check the version of ubuntu in terminal, \$ cat etc/lsb-release

```
DISTRIB_ID=Ubuntu
DISTRIB_RELEASE=24.04
DISTRIB_CODENAME=noble
DISTRIB_DESCRIPTION="Ubuntu 24.04.4 LTS"
```

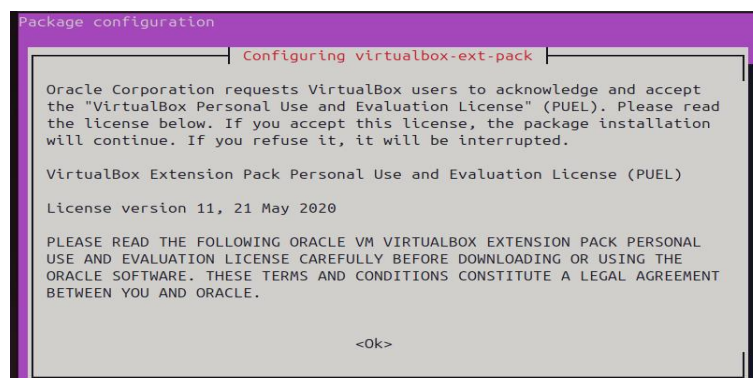
Open terminal in Ubuntu VM and in bash terminal, run:

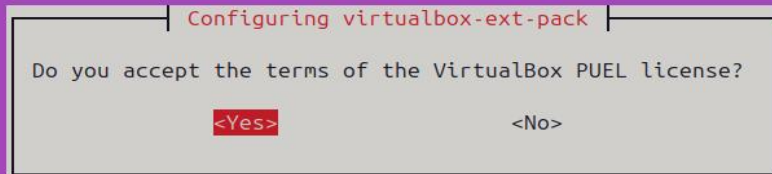
sudo apt update

```
nit6@nit6:~$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
fd0          2:0    1    1.4M  0 disk
loop0        7:0    0     4K  1 loop /snap/bare/5
loop1        7:1    0    74M  1 loop /snap/core22/2292
loop2        7:2    0 251.7M  1 loop /snap/firefox/7766
loop3        7:3    0   18.5M  1 loop /snap/firmware-updater/210
loop4        7:4    0 531.4M  1 loop /snap/gnome-42-2204/247
loop5        7:5    0   91.7M  1 loop /snap/gtk-common-themes/1535
loop6        7:6    0   10.8M  1 loop /snap/snap-store/1270
loop7        7:7    0   48.1M  1 loop /snap/snapd/25935
loop8        7:8    0    576K  1 loop /snap/snapd-desktop-integration/343
sda          8:0    0   120G  0 disk
├─sda1       8:1    0     1M  0 part
└─sda2       8:2    0   120G  0 part /
sr0         11:0    1   99.5M  0 rom  /media/nit6/CDROM
sr1         11:1    1  1024M  0 rom

nit6@nit6:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
nit6@nit6:~$ sudo apt update
[sudo] password for nit6:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
```

sudo apt install -y virtualbox virtualbox-ext-pack





```
nit6@nit6:~$ sudo apt install -y virtualbox-ext-pack
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  virtualbox-ext-pack
0 upgraded, 1 newly installed, 0 to remove and 99 not upgraded.
Need to get 12.3 kB of archives.
After this operation, 165 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 virtualbox-ext-pack all 7.0.16-1 [12.3 kB]
Fetched 12.3 kB in 0s (122 kB/s)
Preconfiguring packages ...
Selecting previously unselected package virtualbox-ext-pack.
(Reading database ... 155346 files and directories currently installed.)
Preparing to unpack ../virtualbox-ext-pack_7.0.16-1_all.deb ...
License has already been accepted.
Unpacking virtualbox-ext-pack (7.0.16-1) ...
Setting up virtualbox-ext-pack (7.0.16-1) ...
virtualbox-ext-pack: downloading: https://download.virtualbox.org/virtualbox/7.0.16/Oracle_VM_VirtualBox_Extension_Pack-7.0.16.vbox-extpack
The file will be downloaded into /var/cache/virtualbox-ext-pack
License accepted.
0%...10%...20%...30%...40%...50%...60%...70%...80%...90%...100%
Successfully installed "Oracle VM VirtualBox Extension Pack".
```

Add HashiCorp key and repo

```
wget -O- https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-archive-keyring.gpg
```

```
echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] \
https://apt.releases.hashicorp.com $(lsb_release -cs) main" | \
sudo tee /etc/apt/sources.list.d/hashicorp.list
```

#Install curl and vagrant through cmd

```
sudo apt install curl
```

```
curl -O https://releases.hashicorp.com/vagrant/2.4.9/vagrant_2.4.9-1_amd64.deb
```

```
sudo apt install ./vagrant_2.4.9-1_amd64.deb
```

```
vagrant --version
```

```
libcurl3t64-gnutls libcurl4t64
Installing:
  apt-transport-https curl
Summary:
  Upgrading: 2, Installing: 2, Removing: 0, Not Upgrading: 24
  Download size: 1,117 kB
  Space needed: 558 kB / 11.0 GB available
0% [waiting for headers]
Ign:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 apt-transport-https all 3.2.0
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libcurl4t64 amd64 8.18.0-1ubuntu2.1 [425 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 curl amd64 8.18.0-1ubuntu2.1 [272 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libcurl3t64-gnutls amd64 8.18.0-1ubuntu2.1 [416 kB]
Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 apt-transport-https all 3.2.0 [3,928 B]
Fetched 1,117 kB in 1min 35s (11.7 kB/s)
Selecting previously unselected package apt-transport-https.
```



```

nit6@nit6:~$ curl -O https://releases.hashicorp.com/vagrant/2.4.9/vagrant_2.4.9-1_amd64.deb
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 86.3M  100 86.3M    0     0  2713k      0  0:00:32  0:00:32 --:--:-- 3431k
nit6@nit6:~$ sudo apt install ./vagrant_2.4.9-1_amd64.deb
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'vagrant' instead of './vagrant_2.4.9-1_amd64.deb'
The following NEW packages will be installed:
  vagrant
0 upgraded, 1 newly installed, 0 to remove and 99 not upgraded.
Need to get 0 B/90.6 MB of archives.
After this operation, 274 MB of additional disk space will be used.
Get:1 /home/nit6/vagrant_2.4.9-1_amd64.deb vagrant amd64 2.4.9-1 [90.6 MB]
Selecting previously unselected package vagrant.
(Reading database ... 155359 files and directories currently installed.)
Preparing to unpack .../nit6/vagrant_2.4.9-1_amd64.deb ...
Unpacking vagrant (2.4.9-1) ...
Setting up vagrant (2.4.9-1) ...
N: Download is performed unsandboxed as root as file '/home/nit6/vagrant_2.4.9-1_amd64.deb' couldn't be accessed by user '_apt'. - pkgAcquire::Run (13: Permission denied)
nit6@nit6:~$ vagrant --version
Vagrant 2.4.9

```

virtualbox --help

```

nit6@nit6:~$ virtualbox --help
Oracle VM VirtualBox VM Selector v7.0.16_Ubuntu
Copyright (C) 2005-2026 Oracle and/or its affiliates

No special options.

If you are looking for --startvm and related options, you need to use VirtualBox VM.
nit6@nit6:~$ mkdir ~/dhcp-lab && cd ~/dhcp-lab

```

Create Lab Folder dhcp-lab

Note: cd ~/dhcp-lab - inside this file vagrant file was installed

Inside Vagrantfile (YAML)

```

# Vagrantfile
Vagrant.configure("2") do |config|

# --- DHCP Server: G1 ---
config.vm.define "G1" do |g1|
  g1.vm.box = "generic/centos8"
  g1.vm.hostname = "G1"

# Static IP on host-only network
g1.vm.network "private_network", ip: "192.168.10.1",
  netmask: "255.255.255.0"

g1.vm.provider "vmware_desktop" do |v|
  v.memory = 1024
  v.cpus = 1
end

# Provision: install & configure DHCP
g1.vm.provision "shell", inline: <<-SHELL
# 1. Install DHCP server

```

```

yum install -y dhcp-server

# 2. Write the DHCP config
cat > /etc/dhcp/dhcpd.conf <<'EOF'
default-lease-time 600;
max-lease-time 7200;

subnet 192.168.10.0 netmask 255.255.255.0 {
    range 192.168.10.1 192.168.10.50;
    option routers 192.168.10.1;
    option domain-name-servers 8.8.8.8;
}
EOF

# 3. Enable & start the service
systemctl enable dhcpd
systemctl start dhcpd
systemctl status dhcpd
SHELL
end

# — Client 1: M1 —————
config.vm.define "M1" do |m1|
  m1.vm.box = "generic/centos8"
  m1.vm.hostname = "M1"

  # DHCP — gets IP from G1
  m1.vm.network "private_network", type: "dhcp"

  m1.vm.provider "vmware_desktop" do |v|
    v.memory = 512
    v.cpus = 1
  end
end

# — Client 2: M2 —————
config.vm.define "M2" do |m2|
  m2.vm.box = "generic/centos8"
  m2.vm.hostname = "M2"
  # DHCP — gets IP from G1
  m2.vm.network "private_network", type: "dhcp"

  m2.vm.provider "vmware_desktop" do |v|
    v.memory = 512
    v.cpus = 1
  end
end
end

```

```

nit6@nit6:~/dhcp-lab$ nano Vagrantfile
nit6@nit6:~/dhcp-lab$ cd ~/dhcp-lab
nit6@nit6:~/dhcp-lab$ vagrant up
Bringing machine 'G1' up with 'virtualbox' provider...
Bringing machine 'M1' up with 'virtualbox' provider...
Bringing machine 'M2' up with 'virtualbox' provider...
==> G1: Box 'generic/centos8' could not be found. Attempting to find and install
...
G1: Box Provider: virtualbox
G1: Box Version: >= 0
==> G1: Loading metadata for box 'generic/centos8'
G1: URL: https://vagrantcloud.com/api/v2/vagrant/generic/centos8
==> G1: Adding box 'generic/centos8' (v4.3.12) for provider: virtualbox (amd64)
G1: Downloading: https://vagrantcloud.com/generic/boxes/centos8/versions/4.3
.12/providers/virtualbox/amd64/vagrant.box
Progress: 13% (Rate: 3196k/s, Estimated time remaining: 0:05:02)

```

Scenario 1: LXC Container Setup

Three LXC containers were launched using CentOS 9-Stream images. G1 acts as the DHCP and DNS server, while M1 and M2 are client machines.

sudo snap install lxd

```
nit6@nit6:~/dhcp-lab$ cd ~
nit6@nit6:~$ sudo snap install lxd
[sudo] password for nit6:
lxd (5.21/stable) 5.21.4-aee7e08 from Canonical✓ installed
nit6@nit6:~$
```

sudo lxd init --auto

sudo usermod -aG lxd --auto

newgrp lxd

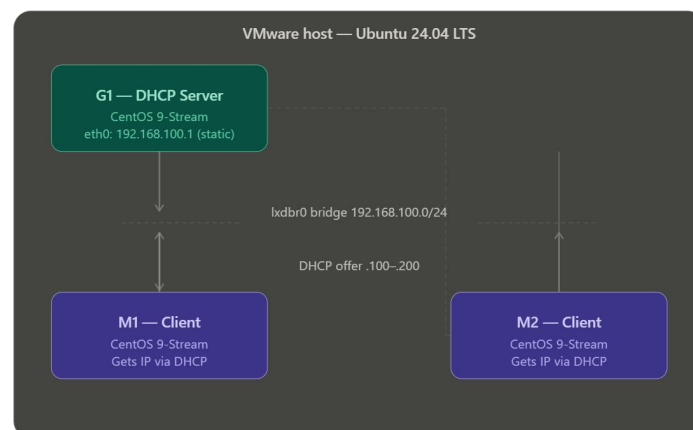
lxc list (Should show empty table ✓)

```
nit6@nit6:~$ sudo lxd init --auto

nit6@nit6:~$
nit6@nit6:~$
nit6@nit6:~$ sudo usermod -aG lxd $USER
nit6@nit6:~$ newgrp lxd
nit6@nit6:~$ lxc list
To start your first container, try: lxc launch ubuntu:24.04
Or for a virtual machine: lxc launch ubuntu:24.04 --vm

+-----+-----+-----+-----+-----+-----+
| NAME | STATE | IPV4 | IPV6 | TYPE | SNAPSHOTS |
+-----+-----+-----+-----+-----+-----+
nit6@nit6:~$
```

1.1 Container Launch Commands



```
lxc launch images:centos/9-Stream G1
lxc launch images:centos/9-Stream M1
lxc launch images:centos/9-Stream M2
```

```
lxc launch images:centos/9-Stream G1
lxc launch images:centos/9-Stream M1
lxc launch images:centos/9-Stream M2
```

1.2 DHCP Server Configuration on G1

Step 1 — Install DHCP Server

```
lxc exec G1 -- bash
yum install -y dhcp-server
```

```
nit6@nit6:~$ lxc exec G1 -- bash
[root@g1 ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host proto kernel_lo
        valid_lft forever preferred_lft forever
4: eth0@if5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:16:3e:79:e9:d6 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.201.2.198/24 brd 10.201.2.255 scope global dynamic noprefixroute eth0
        valid_lft 3358sec preferred_lft 3358sec
    inet6 fd42:7e5:68af:e217:f42a:b1bf:f0b0:c405/64 scope global noprefixroute
        valid_lft forever preferred_lft forever
    inet6 fe80::8653:388:e8ea:521b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@g1 ~]#
```

```
Install 1 Package

Total download size: 1.2 M
Installed size: 3.9 M
Downloading Packages:
dhcp-server-4.4.2-20.b1.el9.x86_64.rpm                                677 kB/s | 1.2 MB    00:01
-----
Total                                                                609 kB/s | 1.2 MB    00:02
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                                    1/1
  Running scriptlet: dhcp-server-12:4.4.2-20.b1.el9.x86_64          1/1
  Installing     : dhcp-server-12:4.4.2-20.b1.el9.x86_64          1/1
  Running scriptlet: dhcp-server-12:4.4.2-20.b1.el9.x86_64          1/1
  Verifying      : dhcp-server-12:4.4.2-20.b1.el9.x86_64          1/1

Installed:
  dhcp-server-12:4.4.2-20.b1.el9.x86_64

Complete!
[root@g1 ~]#
```

Step 2 — Configure /etc/dhcp/dhcpd.conf

```
default-lease-time 600;
max-lease-time 7200;
subnet 192.168.10.0 netmask 255.255.255.0 {
    range 192.168.10.1 192.168.10.50;
    option routers 192.168.10.1;
    option domain-name-servers 8.8.8.8;
}
```



```
nitubuntu@nitubuntu-VMware-Virtual-Platform: ~ — sudo nano /etc/dhcp/dhcpd.conf
GNU nano 8.7.1 /etc/dhcp/dhcpd.conf *
# dhcpd.conf
#
# Sample configuration file for ISC dhcpd
#
# Attention: If /etc/ltsp/dhcpd.conf exists, that will be used as
# configuration file instead of this file.
#
# option definitions common to all supported networks...
option domain-name "example.org";
option domain-name-servers ns1.example.org, ns2.example.org;

subnet 192.168.1.0 netmask 255.255.255.0 {
    range 192.168.1.100 192.168.1.200;
    option routers 192.168.1.1;
    option domain-name-servers 8.8.8.8, 8.8.4.4;
    default-lease-time 600;
    max-lease-time 7200;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line

ct input to this VM, click inside or press Ctrl+G.
```

Step 3 — Enable and Start DHCP

```
systemctl enable dhcpd
systemctl start dhcpd
systemctl status dhcpd
```

```
Complete!
[vagrant@G1 ~]$ sudo systemctl daemon-reload
Unknown operation daemon-reload.
[vagrant@G1 ~]$ sudo systemctl unmask dhcpd
[vagrant@G1 ~]$ sudo systemctl enable dhcpd
[vagrant@G1 ~]$ sudo systemctl start dhcpd
[vagrant@G1 ~]$ sudo systemctl status dhcpd
● dhcpd.service - DHCPv4 Server Daemon
   Loaded: loaded (/usr/lib/systemd/system/dhcpd.service; enabled; vendor preset: disabled)
   Active: active (running) since Thu 2026-05-14 11:48:54 UTC; 4s ago
     Docs: man:dhcpd(8)
           man:dhcpd.conf(5)
  Main PID: 3829 (dhcpd)
    Status: "Dispatching packets..."
     Tasks: 1 (limit: 4952)
    Memory: 8.6M
    CGroup: /system.slice/dhcpd.service
            └─3829 /usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd -

May 14 11:48:54 G1 dhcpd[3829]:
May 14 11:48:54 G1 dhcpd[3829]: No subnet declaration for eth0 (10.0.2.15).
May 14 11:48:54 G1 dhcpd[3829]: ** Ignoring requests on eth0. If this is not what
May 14 11:48:54 G1 dhcpd[3829]: you want, please write a subnet declaration
May 14 11:48:54 G1 dhcpd[3829]: in your dhcpd.conf file for the network segment
May 14 11:48:54 G1 dhcpd[3829]: to which interface eth0 is attached. **
May 14 11:48:54 G1 dhcpd[3829]:
May 14 11:48:54 G1 dhcpd[3829]: Sending on Socket/fallback/fallback-net
May 14 11:48:54 G1 dhcpd[3829]: Server starting service.
May 14 11:48:54 G1 systemd[1]: Started DHCPv4 Server Daemon.
[vagrant@G1 ~]$
```

1.3 Static IP Configuration on G1

Note: In my system (wired connection 1) was the name mentioned to the system

```
==> M2: flag to force provisioning. Provisioners marked to run always w
nit6@nit6:~/dhcp-lab$ vagrant status
Current machine states:

G1               running (virtualbox)
M1               running (virtualbox)
M2               running (virtualbox)

This environment represents multiple VMs. The VMs are all listed
```

All 3 Containers are Running! ✓

```
nit6@nit6:~$
nit6@nit6:~$ lxc list
+-----+-----+-----+-----+-----+-----+
| NAME | STATE | IPV4 | IPV6 |
| TYPE | SNAPSHOTS |
+-----+-----+-----+-----+-----+
| G1 | RUNNING | 10.201.2.198 (eth0) | fd42:7e5:68af:e217:f42a:b1bf:f0b0:c405 (eth0) | CONTAINER | 0 |
+-----+-----+-----+-----+-----+-----+
| M1 | RUNNING | 10.201.2.145 (eth0) | fd42:7e5:68af:e217:b791:48b2:d3f7:25d3 (eth0) | CONTAINER | 0 |
+-----+-----+-----+-----+-----+-----+
| M2 | RUNNING | 10.201.2.74 (eth0) | fd42:7e5:68af:e217:b153:ae00:9470:cf47 (eth0) | CONTAINER | 0 |
+-----+-----+-----+-----+-----+-----+
```

Static IP was configured using nmcli and Netplan:

```
nmcli con mod "Wired connection 1" \
  ipv4.addresses 192.168.100.1/24 \
  ipv4.method manual \
  ipv4.gateway "10.201.2.1" \
  ipv4.dns "8.8.8.8"
nmcli con up "Wired connection 1"
```

Netplan config created for static IP:

```
cat > /etc/netplan/01-static.yaml << 'EOF'
network:
  version: 2
  ethernets:
    eth0:
      dhcp4: no
      addresses:
        - 192.168.100.1/24
EOF
```

Running ip addr and ip a commands :

```
nit6@nit6:~$ lxc exec G1 -- bash -c 'cat > /etc/sysconfig/network-scripts/lo
eth0 << EOF
> DEVICE=eth0
> BOOTPROTO=static
> IPADDR=10.201.2.100
> NETMASK=255.255.255.0
> GATEWAY=10.201.2.1
> DNS1=8.8.8.8
> ONBOOT=yes
> EOF'
nit6@nit6:~$ lxc restart G1
nit6@nit6:~$ lxc exec G1 -- ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=127 time=10.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=127 time=8.28 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=127 time=8.15 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 8.153/9.069/10.781/1.211 ms
nit6@nit6:~$
```

```
nit6@nit6:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1
000
    link/ether 00:0c:29:fc:17:cd brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.245.138/24 brd 192.168.245.255 scope global dynamic noprefixroute ens33
        valid_lft 1000sec preferred_lft 1000sec
    inet6 fe80::20c:29ff:fefc:17cd/64 scope link
        valid_lft forever preferred_lft forever
3: vboxnet0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:00 brd ff:ff:ff:ff:ff:ff
4: vboxnet1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 0a:00:27:00:00:01 brd ff:ff:ff:ff:ff:ff
5: lxdbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1
000
    link/ether 00:16:3e:d2:0b:11 brd ff:ff:ff:ff:ff:ff
    inet 10.201.2.1/24 scope global lxdbr0
        valid_lft forever preferred_lft forever
    inet6 fd42:7e5:68af:e217::1/64 scope global
        valid_lft forever preferred_lft forever
    inet6 fe80::216:3eff:fed2:b11/64 scope link
        valid_lft forever preferred_lft forever
```



```
nit6@nit6: ~/dhcp-lab
M1: prevent things such as shared folders from working properly. If you see
M1: shared folder errors, please make sure the guest additions within the
M1: virtual machine match the version of VirtualBox you have installed on
M1: your host and reload your VM.
M1:
M1: Guest Additions Version: 6.1.30 r148432
M1: VirtualBox Version: 7.0
==> M1: Setting hostname...
==> M1: Configuring and enabling network interfaces...
==> M1: Machine already provisioned. Run `vagrant provision` or use the `--provision`
==> M1: flag to force provisioning. Provisioners marked to run always will still run.
==> M2: Checking if box 'generic/centos8' version '4.3.12' is up to date...
==> M2: Clearing any previously set forwarded ports...
==> M2: Fixed port collision for 22 => 2222. Now on port 2201.
==> M2: Clearing any previously set network interfaces...
==> M2: Preparing network interfaces based on configuration...
M2: Adapter 1: nat
M2: Adapter 2: hostonly
==> M2: Forwarding ports...
M2: 22 (guest) => 2201 (host) (adapter 1)
==> M2: Running 'pre-boot' VM customizations...
==> M2: Booting VM...
==> M2: Waiting for machine to boot. This may take a few minutes...
M2: SSH address: 127.0.0.1:2201
M2: SSH username: vagrant
M2: SSH auth method: private key
==> M2: Machine booted and ready!
==> M2: Checking for guest additions in VM...
M2: The guest additions on this VM do not match the installed version of
M2: VirtualBox! In most cases this is fine, but in rare cases it can
```



1.5 Verification

- `systemctl status dhcpd` — confirmed active and running on G1
- M1 received dynamic IP from G1 DHCP server
- M2 received dynamic IP from G1 DHCP server
- G1 ping to 8.8.8.8 — 0% packet loss, internet connectivity confirmed

FOR EXPECTED OUTPUT

Step 1 — Install net-tools on M1 (for ifconfig):

bash

vagrant ssh M1 -c "sudo yum install -y net-tools"

```
nit6@nit6:~/dhcp-lab$ vagrant ssh M1 -c "sudo yum install -y net-tools"
CentOS Linux 8 - AppStream                               6.9 kB/s | 4.3 kB
CentOS Linux 8 - BaseOS                                  51 kB/s | 3.9 kB
CentOS Linux 8 - Extras                                  22 kB/s | 1.5 kB
Extra Packages for Enterprise Linux 8 - x86_64           3.8 kB/s | 9.1 kB
Package net-tools-2.0-0.52.20160912git.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
nit6@nit6:~/dhcp-lab$
```

Step 2 — Run ifconfig on M1:

bash

vagrant ssh M1 -c "ifconfig"

```
nit6@nit6:~/dhcp-lab$ vagrant ssh M1 -c "ifconfig"
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
    ether 08:00:27:b7:f6:78  txqueuelen 1000  (Ethernet)
    RX packets 807  bytes 133502 (130.3 KiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 644  bytes 107802 (105.2 KiB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.10.11  netmask 255.255.255.0  broadcast 192.168.10.255
    inet6 fe80::a00:27ff:fe5f:7ed1  prefixlen 64  scopeid 0x20<link>
    ether 08:00:27:5f:7e:d1  txqueuelen 1000  (Ethernet)
    RX packets 6  bytes 360 (360.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 18  bytes 1296 (1.2 KiB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    loop txqueuelen 1000  (Local Loopback)
    RX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 0  bytes 0 (0.0 B)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

nit6@nit6:~/dhcp-lab$
```

Step 3 — Install net-tools on M2:

vagrant ssh M2 -c "sudo yum install -y net-tools"

```
nit6@nit6:~/dhcp-lab$ vagrant ssh M2 -c "sudo yum install -y net-tools"
CentOS Linux 8 - AppStream                               6.6 kB/s | 4.3 kB
CentOS Linux 8 - BaseOS                                  41 kB/s | 3.9 kB
CentOS Linux 8 - Extras                                  16 kB/s | 1.5 kB
Extra Packages for Enterprise Linux 8 - x86_64           2.1 MB/s | 14 MB

Package net-tools-2.0-0.52.20160912git.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

Step 4 — Run ifconfig on M2:

bash

vagrant ssh M2 -c "ifconfig"

```
nit6@nit6:~/dhcp-lab$ vagrant ssh M2 -c "ifconfig"
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    ether 08:00:27:b7:f6:78 txqueuelen 1000 (Ethernet)
    RX packets 11699 bytes 15832924 (15.0 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1993 bytes 190817 (186.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.10.12 netmask 255.255.255.0 broadcast 192.168.10.255
    inet6 fe80::a00:27ff:fe52:6b8c prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:52:6b:8c txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 18 bytes 1296 (1.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

For VirtualBox GUI login,

